Lecture 12 (10/12)

Study design, confounding, and multiple predictors

Unit 3 summary:

- An estimator $\hat{\theta}$ is how we approximate a parameter θ from data. Its variation over replicates of the data (repeated experiments, different samples) is the sample distribution.
- The error breaks down into bias and variance, $\text{MSE}_{\hat{\theta}} = \text{se}(\hat{\theta})^2 + \text{Bias}^2$. We look for consistent estimators where both vanish.
- Confidence intervals and p-values are computed from the sample distribution.
- The least squares estimators $\hat{\beta}_0, \hat{\beta}_1$ provide unbiased estimates of parameters in a regression model (assuming data points are sampled independently). $R^2 = \hat{\rho}^2$ is the estimate of ρ^2.
- Autoregressive models (specifically AR(1) models) are a type of regression model where the next value in the time series is the response variable. Here the regression coefficient is the correlation and as ρ approaches 1 the long-term variance diverges
- In AR(1) models, the (X, Y) points are successive values (Y_t, Y_{t+1}) and the pairs are therefore not independent, leading to a situation where least squares estimators are biased (you don't have to know the formula, but understand why it is biased).

Unit 4: regression with multiple predictors.

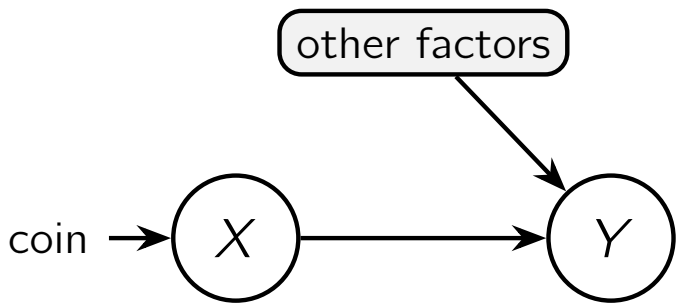
- 4.1 Study design and confounding (today – briefly).
- 4.2 Multiple-predictor regression and its interpretation (today).
- 4.3–4.4 Estimating the coefficients, their sample distribution, collinearity.
- 4.5 Categorical predictors.

Two ways to collect data

Does a treatment ($X = 1$) change an outcome Y, compared with no treatment ($X = 0$)?

Randomized controlled trial (RCT)

we assign X by a coin flip



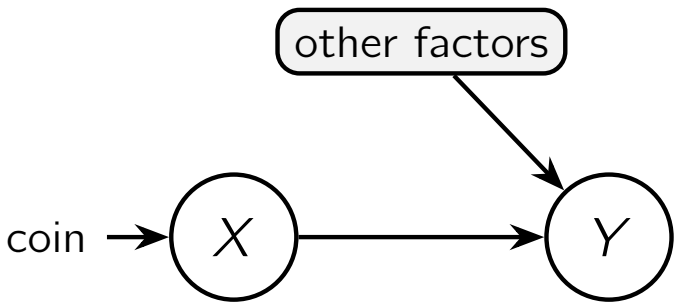
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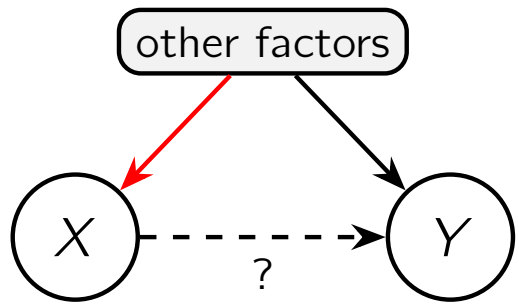
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Cohort study (observational)

we follow people and observe who gets X



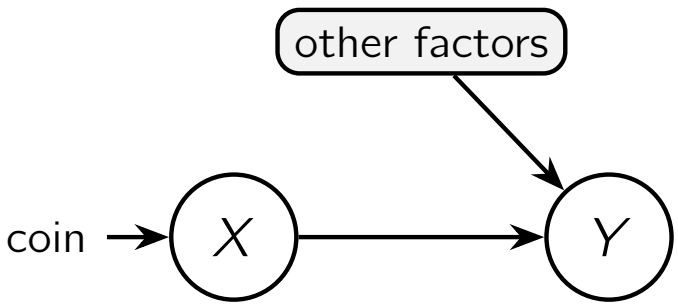
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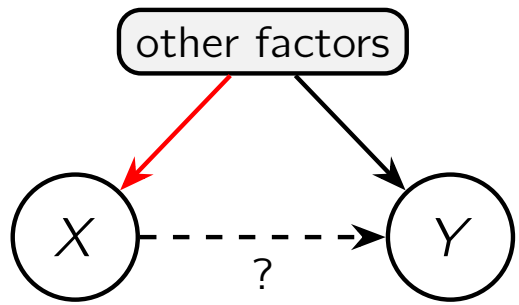
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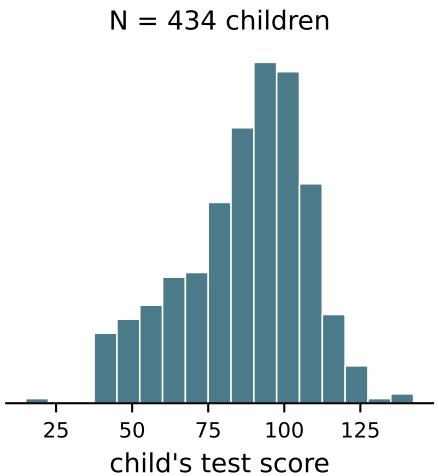
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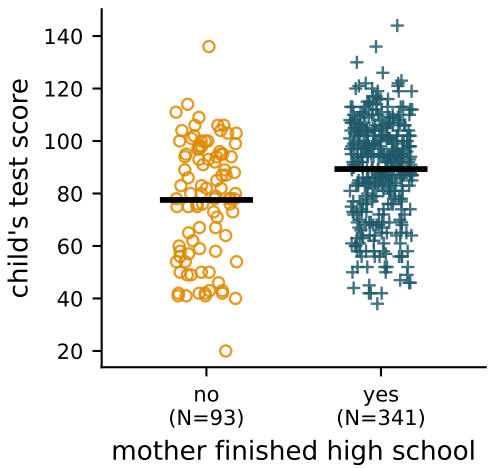
For most questions in the social sciences, an RCT is impossible.

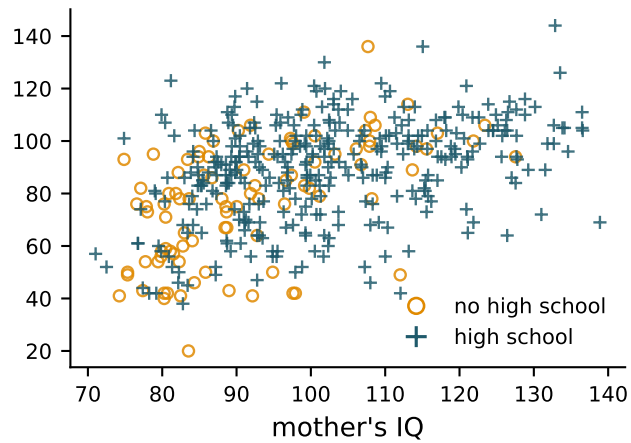
Children's test scores

Test scores of $N = 434$ children, with two facts about each child's mother:

- Y: the child's test score;
- $X_{\text{hs}} = 1$ if the mother finished high school, 0 if not;
- X_{iq}: the mother's IQ.[1]





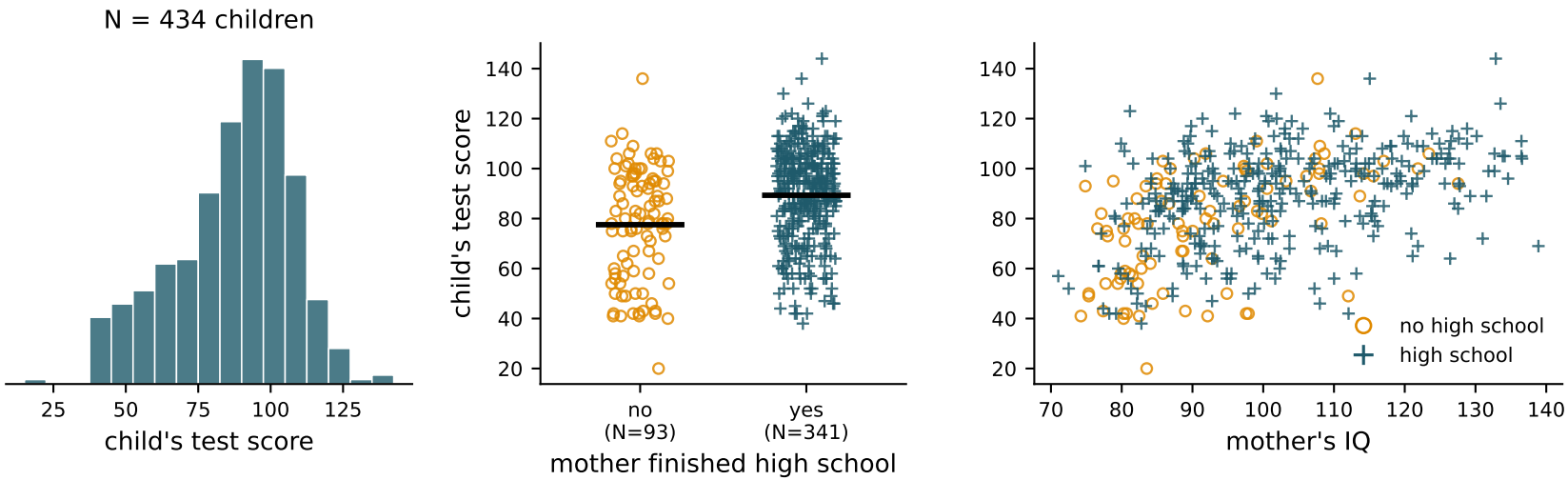


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Example

From the right two plots, estimate σ_{IQ}, σ_{HS}, σ_Y as well as the regression coefficients and correlation coefficients.

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Children's test scores

Standard deviations ($N = 434$):

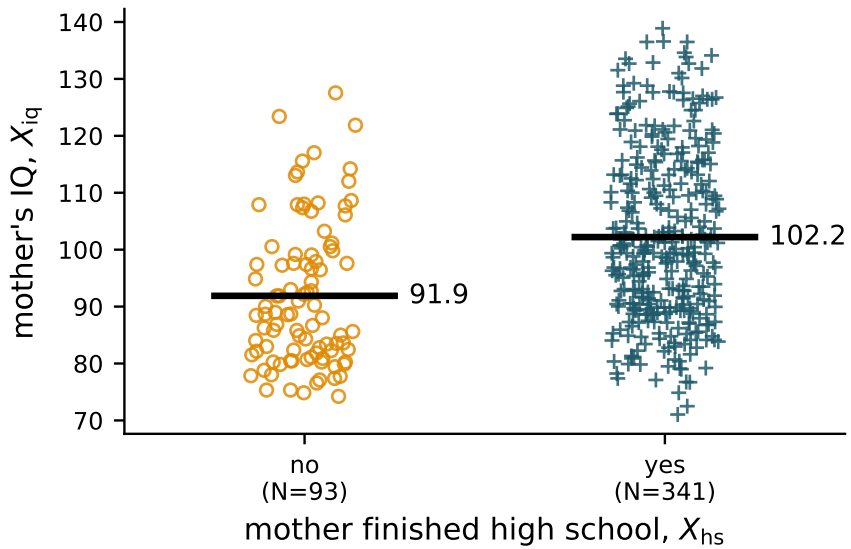
	standard deviation
Y (child's score)	$\sigma_Y = 20.4$
X_{hs}	$\sigma_{\text{HS}} = 0.41$
X_{iq}	$\sigma_{\text{IQ}} = 15$

Single-predictor regressions. We write β' (with a prime) for a single-predictor coefficient, to distinguish it from the coefficients β of the two-predictor model we will fit later:

- Y on X_{hs}: $\hat{\beta}'_{\text{hs}} = 11.8$ (the difference in average score between the two groups), $\hat{\rho} = \hat{\beta}'_{\text{hs}}\sigma_{\text{HS}}/\sigma_Y \approx 0.24$.
- Y on X_{iq}: $\hat{\beta}'_{\text{iq}} = 0.61$, $\hat{\rho} = \hat{\beta}'_{\text{iq}}\sigma_{\text{IQ}}/\sigma_Y \approx 0.45$.

IQ and HS

Mother's IQ vs. whether she finished high school:



What we computed vs. what we want

- **Computed from single predictor regression**: the difference in average score between the groups,

$$\beta'_{\text{hs}} = E[Y \mid X_{\text{hs}} = 1] - E[Y \mid X_{\text{hs}} = 0] \qquad (\hat{\beta}'_{\text{hs}} = 11.8).$$

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- To compare children whose mothers have the *same* IQ, put both variables in the model:

$$Y = \beta_0 + \beta_{\text{hs}} X_{\text{hs}} + \beta_{\text{iq}} X_{\text{iq}} + \epsilon, \qquad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2).$$

Fitting the model in Python: `statsmodels`

The `statsmodels` package fits regression models and reports the estimates and their standard errors:

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No. Observations:                 434   Adj. R-squared:                  0.210
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         25.7315      5.875      4.380      0.000      14.184      37.279
mom_hs         5.9501      2.212      2.690      0.007       1.603      10.297
mom_iq         0.5639      0.061      9.309      0.000       0.445       0.683
==============================================================================
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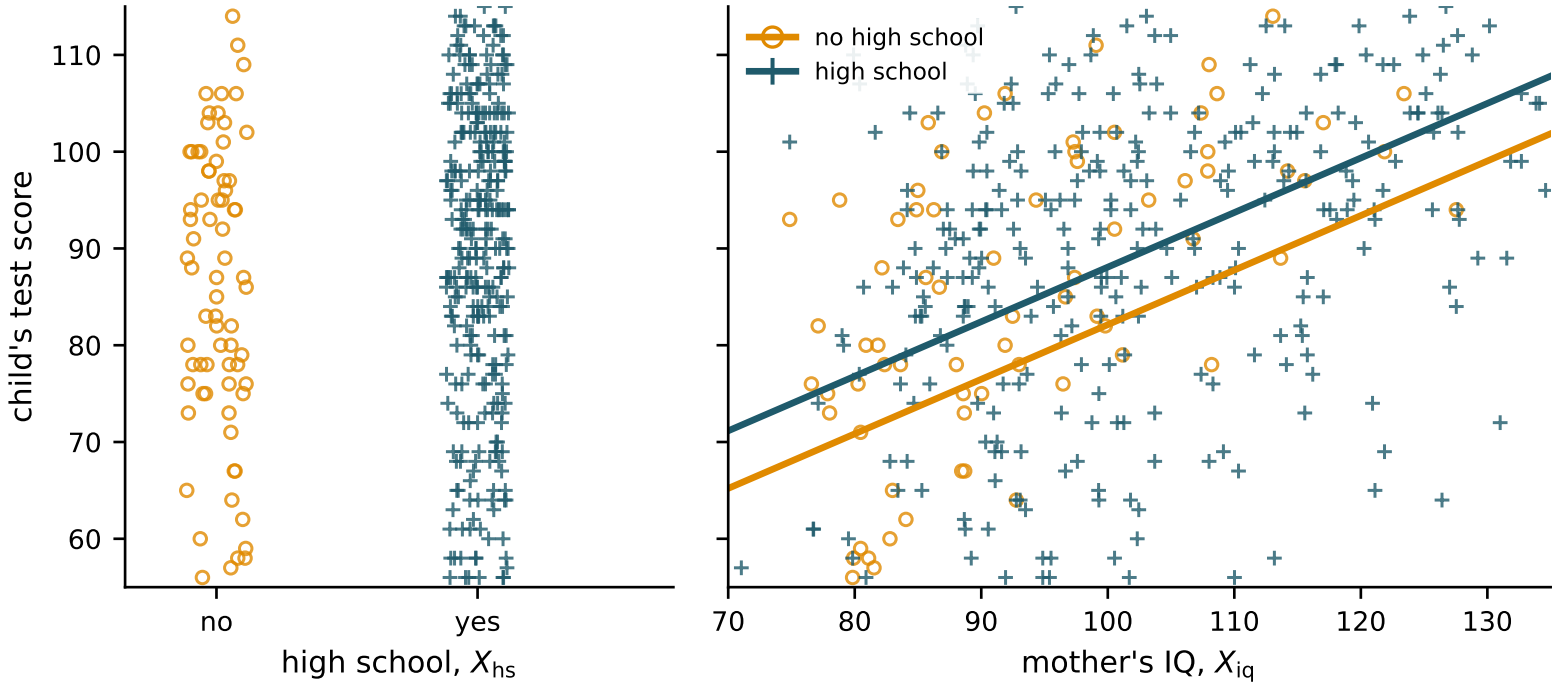
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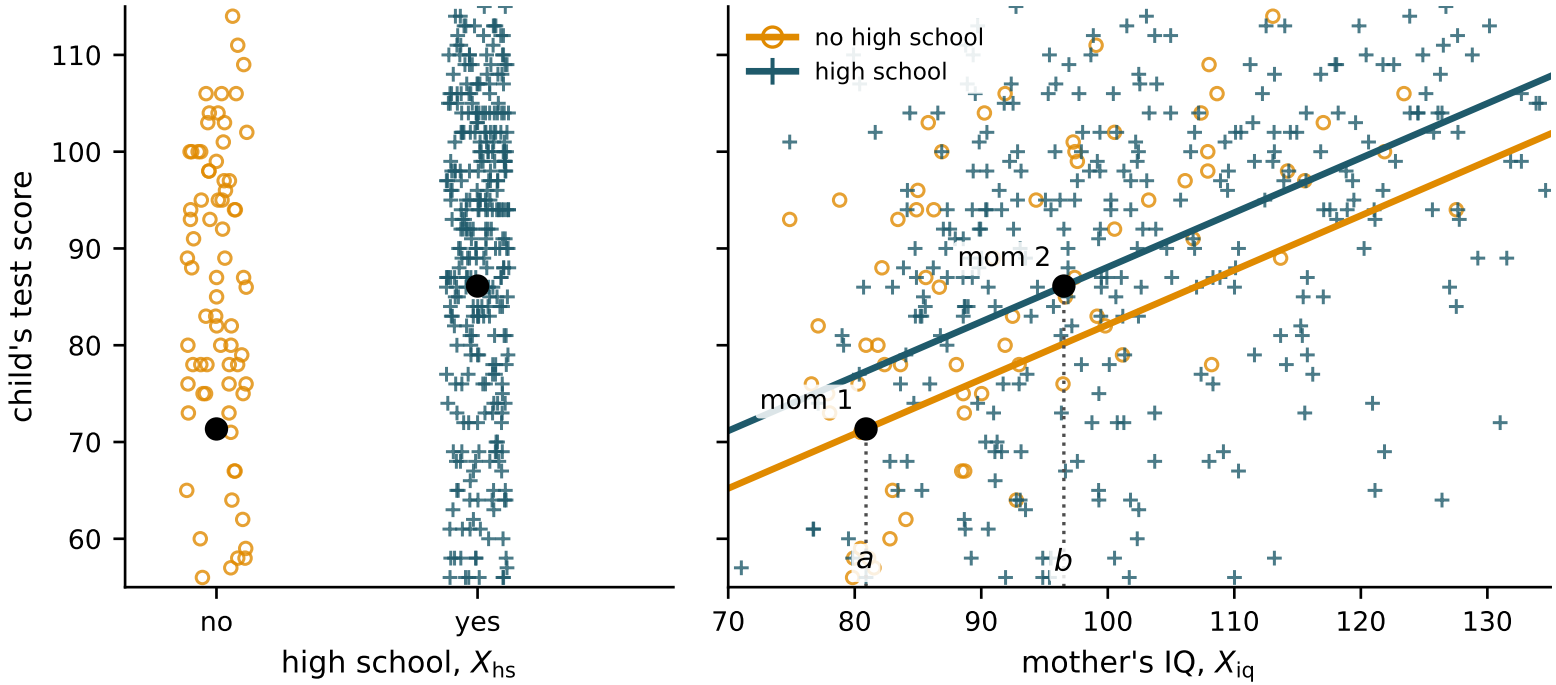
The `coef` column gives the least-squares estimates: $\hat{\beta}_0 = 25.7$, $\hat{\beta}_{\text{hs}} = 5.95$, $\hat{\beta}_{\text{iq}} = 0.56$. (`add_constant` adds the intercept β_0; the other columns come later.)

From two predictors to one



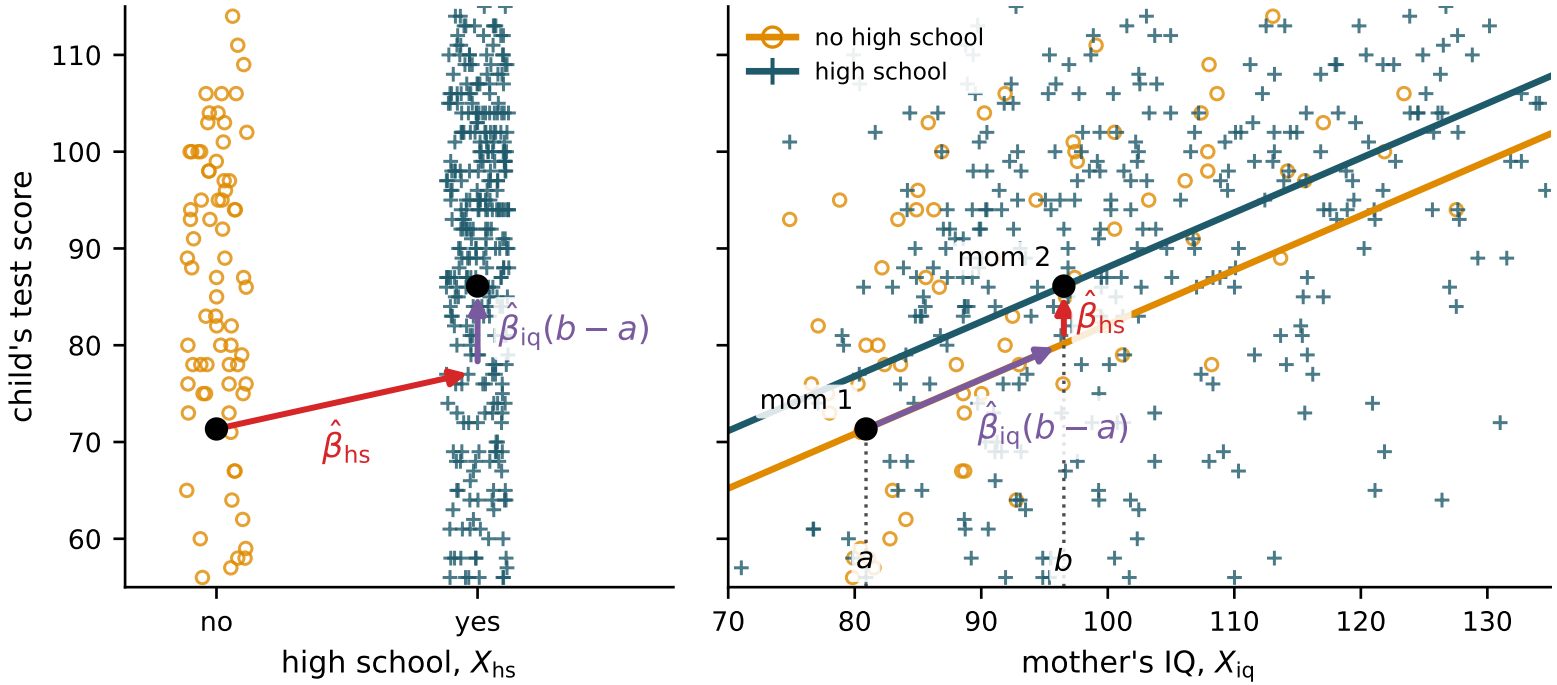
The fitted two-predictor model, vs. schooling (left) and vs. IQ (right).

From two predictors to one



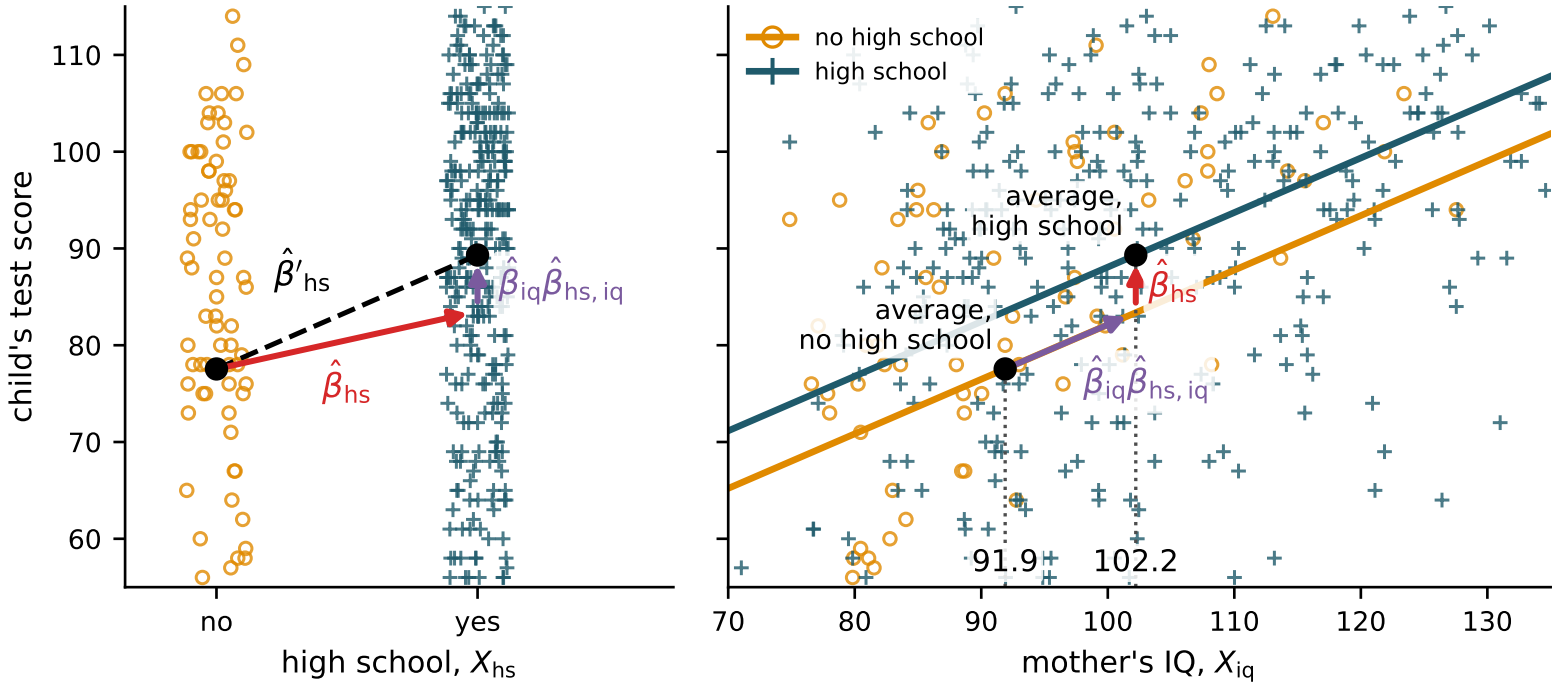
Pick a random mom without high school (IQ $= a$) and one with (IQ $= b$).

From two predictors to one



Their difference in expected score is β_{hs} (schooling, IQ fixed) + $\beta_{\text{iq}}(b-a)$ (difference in IQ).

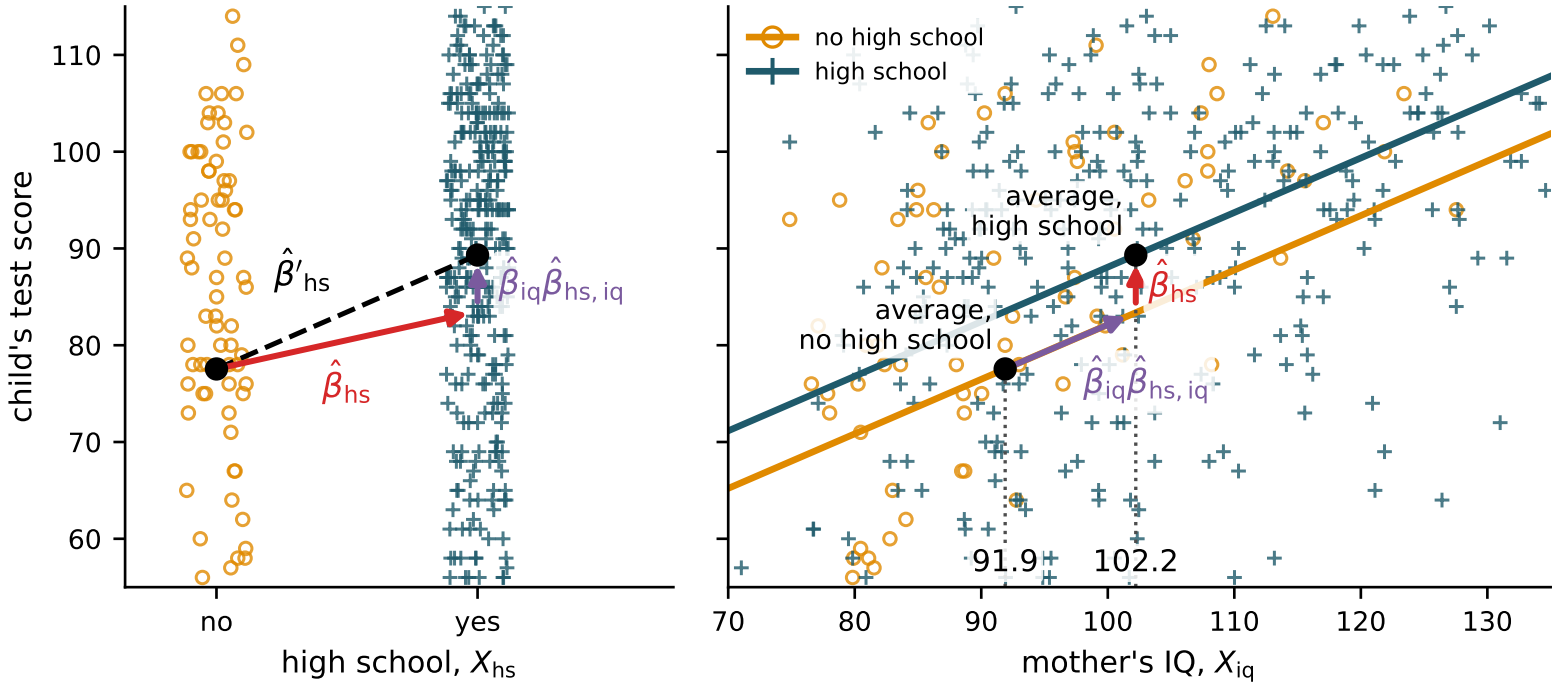
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Average over all moms in each group: a and b are replaced by the average IQs,

$$\underbrace{E[Y \mid X_{\text{hs}} = 1] - E[Y \mid X_{\text{hs}} = 0]}_{\beta'_{\text{hs}}} = \beta_{\text{hs}} + \beta_{\text{iq}} \underbrace{\big(E[X_{\text{iq}} \mid X_{\text{hs}} = 1] - E[X_{\text{iq}} \mid X_{\text{hs}} = 0]\big)}_{\beta_{\text{hs,iq}}}$$

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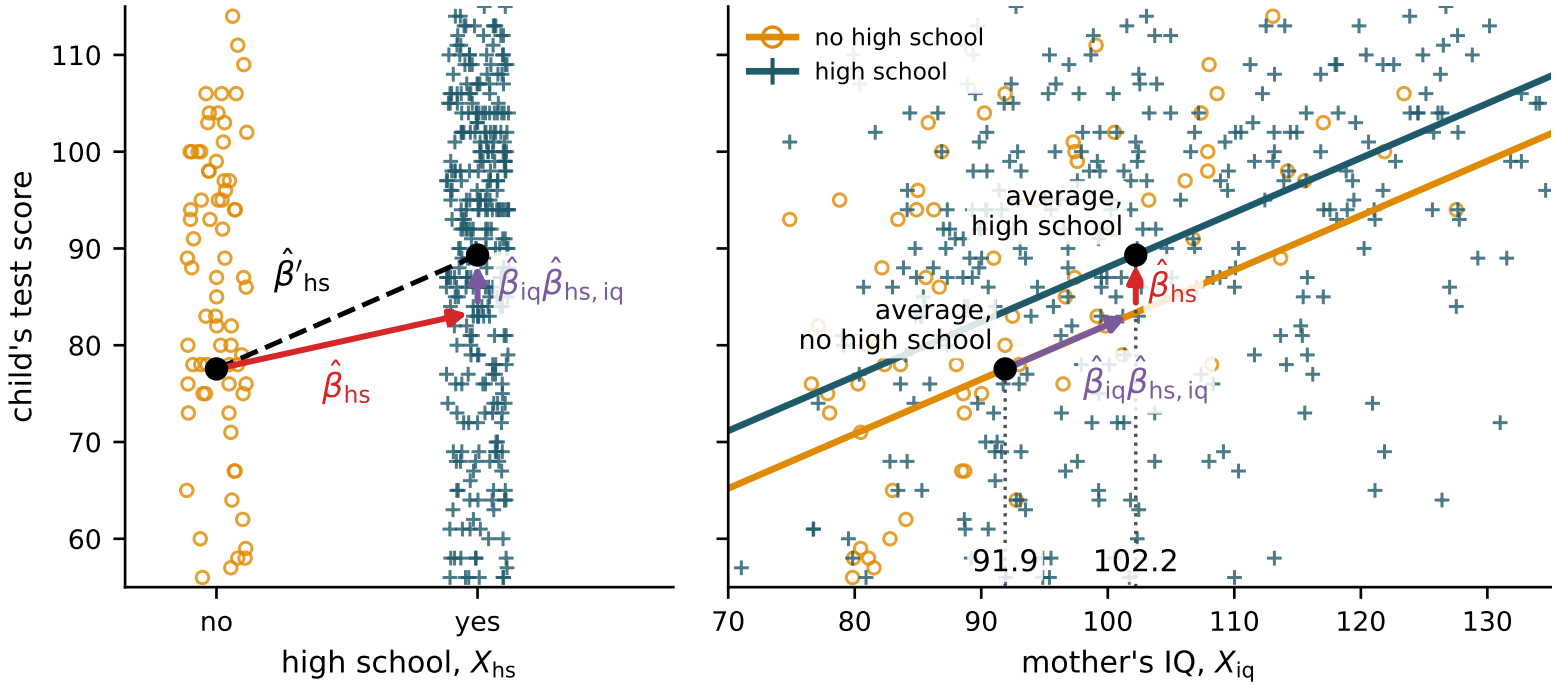


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so $\beta'_{hs} = \beta_{hs} + \beta_{iq}\beta_{hs,iq}$: $11.8 = 5.95 + 0.56 \times 10.3$.

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so $\beta'_{\text{hs}} = \beta_{\text{hs}} + \beta_{\text{iq}}\beta_{\text{hs,iq}}$: $11.8 = 5.95 + 0.56 \times 10.3$.
We say β_{hs} is the effect of schooling controlling for IQ (and vice versa).

Which predictor matters more?

$\hat{\beta}_{\text{hs}} = 5.95$ is ten times $\hat{\beta}_{\text{iq}} = 0.56$, but the coefficients have different units (points for finishing high school vs. points per IQ point), so we cannot compare them directly.

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print(b[1] * np.std(mom_hs))   # high school
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8.449
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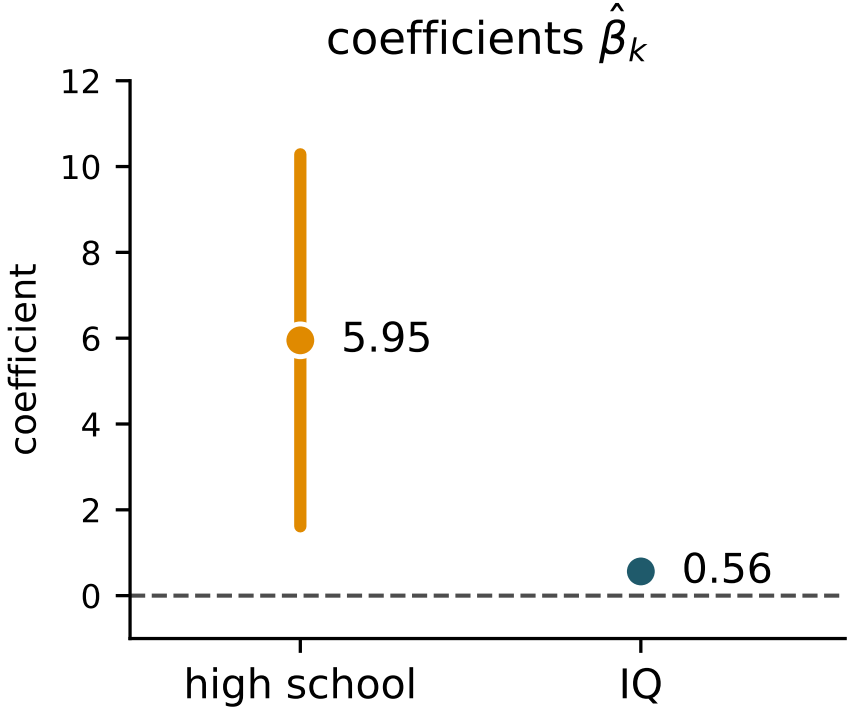
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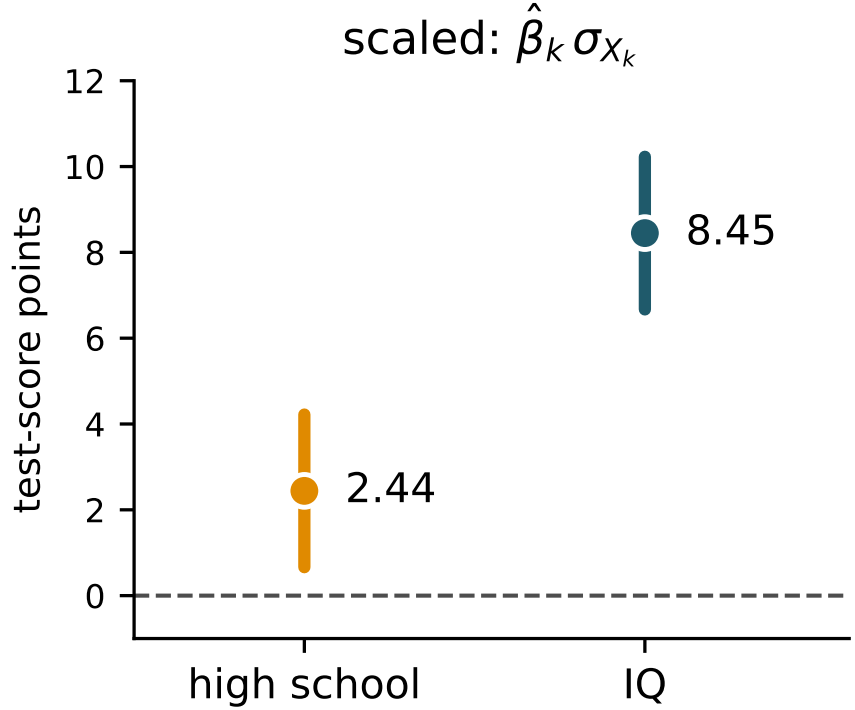
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A one-standard-deviation difference in IQ (15 points) goes with about 8.4 more points on the test; one in schooling ($\sigma_{\text{HS}} = 0.41$) with only about 2.4. IQ matters more.

Raw vs. scaled coefficients

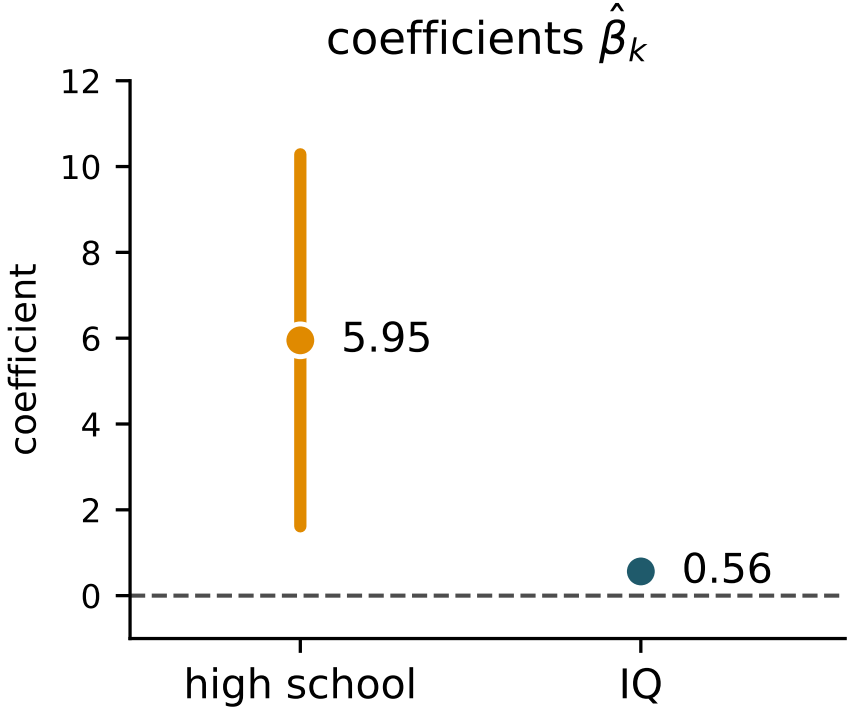
Each coefficient with its 95% confidence interval, before and after multiplying by the standard deviation of its predictor:

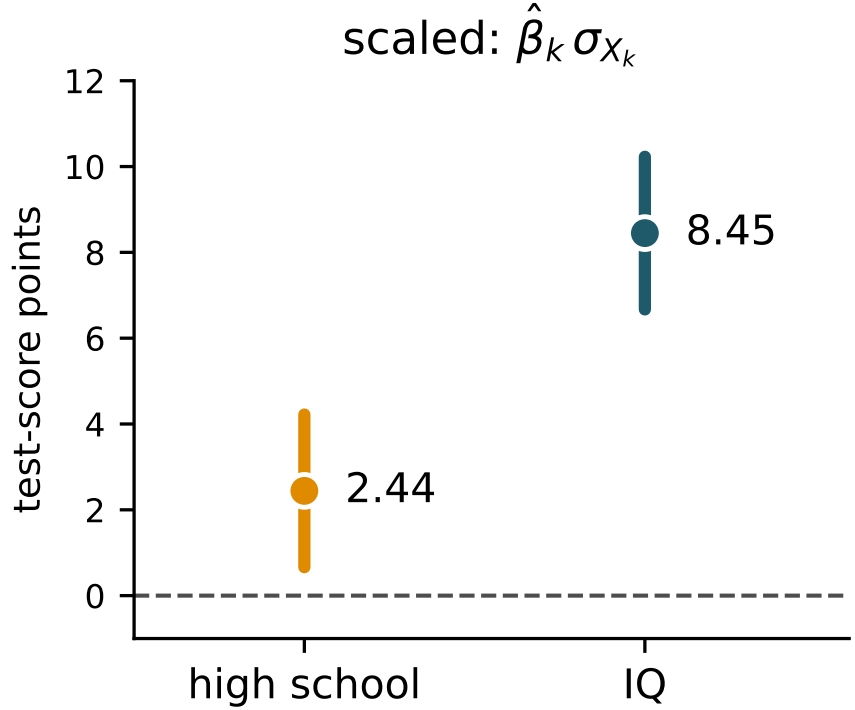




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R^2 with multiple predictors

R^2 is defined exactly as before,

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}}, \qquad \text{SSR} = \sum_{i=1}^{N} (Y_i - \hat{Y}_i)^2, \qquad \text{SST} = \sum_{i=1}^{N} (Y_i - \overline{Y})^2,$$

where now the fitted value $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \cdots + \hat{\beta}_K X_{K,i}$ uses all K predictors ($X_{k,i}$ is the value of predictor k for data point i). It is the fraction of the variation in Y explained by *all* the predictors together:

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`statsmodels` reports it at the top of the summary:

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R^2 for the test scores

$R^2 = 1 - \text{SSR}/\text{SST}$ for three models of the test scores:

predictors	R^2
high school only	0.06
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- About 79% of the variation is *not* explained: the scores have standard deviation 20, and the residuals still have $\hat{\sigma}_\epsilon \approx 18$. The model says little about any child's score.

Example

Midterm 1 scores for $N = 64$ students in this class: y is the score (fraction of points), `difficult` $= 1$ if you rated the difficulty 4 (the highest rating given) in the week 1 survey, and `QSS` $= 1$ for QSS majors.

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.8705      0.033     26.343      0.000       0.804       0.937
difficult     -0.1002      0.040     -2.507      0.015      -0.180      -0.020
QSS           -0.1340      0.040     -3.361      0.001      -0.214      -0.054
==============================================================================
```

```
print(f"{np.std(difficult):.3f}")
print(f"{np.std(QSS):.3f}")
0.498
0.499
```

	difficult		
	0	1	total
QSS $= 0$	0.281	0.188	0.469
QSS $= 1$	0.266	0.266	0.531
total	0.547	0.453	1

Which is the more important predictor of the midterm score? What is the overall regression coefficient (if no other predictors are included) of difficulty?

Solution: midterm scores

More important predictor. Both predictors are 0/1 with standard deviation ≈ 0.5, so scaling barely changes anything:

$$\hat{\beta}_{\text{diff}}\,\sigma_{\text{diff}} = -0.100 \times 0.498 = -0.050, \quad \hat{\beta}_{\text{QSS}}\,\sigma_{\text{QSS}} = -0.134 \times 0.499 = -0.067.$$

QSS is (slightly) more important; here we can compare the raw coefficients directly.

Solution: midterm scores

More important predictor. Both predictors are 0/1 with standard deviation ≈ 0.5, so scaling barely changes anything:

$$\hat{\beta}_{\text{diff}}\,\sigma_{\text{diff}} = -0.100 \times 0.498 = -0.050, \quad \hat{\beta}_{\text{QSS}}\,\sigma_{\text{QSS}} = -0.134 \times 0.499 = -0.067.$$

QSS is (slightly) more important; here we can compare the raw coefficients directly.

Single-predictor coefficient of difficulty. $\beta'_{\text{diff}} = \beta_{\text{diff}} + \beta_{\text{QSS}}\,\beta_{\text{diff,QSS}}$, where from the table

$$\begin{aligned}\hat{\beta}_{\text{diff,QSS}} &= E[X_{\text{QSS}} \mid X_{\text{diff}} = 1] - E[X_{\text{QSS}} \mid X_{\text{diff}} = 0] \\ &= \frac{0.266}{0.453} - \frac{0.266}{0.547} = 0.587 - 0.486 = 0.101,\end{aligned}$$

so $\hat{\beta}'_{\text{diff}} = -0.100 + (-0.134)(0.101) = -0.114$. (Fitting y on `difficult` alone gives -0.1137.)